

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		<u>Angle</u> where (1) accept diagram showing angle arc length equals the radius or approximately 57.3° or $2\pi = 360^\circ$ (1)	2			2		
	(b)		Period, $T = 105 \times 60 = 6300$ [s] or 105 [min] (1) Substitution (regardless of unit): $\omega = \frac{2\pi}{T} = \frac{2\pi}{6300}$ (1) $= 9.97 \times 10^{-4}$ [rad s ⁻¹](1)	1	1 1		3	3	
	(c)		$m\omega^2 R = \frac{GM_E m}{R^2}$ (1) Accept $\frac{mv^2}{r} = \frac{GM_E m}{R^2}$ $R^3 = \frac{GM_E}{\omega^2}$ [so: $R = \sqrt[3]{\frac{GM_E}{\omega^2}}$] (1)		2		2	1	
	(d)		Substitution: $R = \sqrt[3]{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{(9.97 \times 10^{-4})^2}}$ ecf (1) $R = 7\,380$ k[m] (1) Altitude above the surface of the Earth = $7\,380 - 6\,400 = 980$ k[m] (1)	1	1 1		3	3	
	(e)		$R = \sqrt[3]{\frac{(6.67 \times 10^{-11})(7.3 \times 10^{22})}{(9.97 \times 10^{-4})^2}}$ ecf (1) $R = 1\,698$ k[m] (1) Correct conclusion based on candidate's answer e.g. not possible since inside Moon (1) Alternative: Substituting into equation to calculate period or to calculate mass (1) Correct answer e.g. to period (1) Correct conclusion based on period / mass e.g. period longer so not possible, mass smaller so not possible (1)			3	3	1	

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			Alternative: Can also work centripetal acceleration (1.7) (1) Calculate gravitational acceleration (1.6) (1) Not possible (since gravity too weak) (1)						
			Question 4 total	4	6	3	13	8	0